

Suraj Khaire

📍 Pune | ✉️ suraj.khaire01@outlook.com | ☎️ +91-7028748066 | [in](#) [suraj-khaire](#) | [🌐](#) [Suraj-Khaire-01](#)

Education

Vishwakarma Institute of Information Technology

2023 – 2027

B.Tech in Computer Engineering

CGPA: 8.51

Experience

Software Engineering Intern | BMC Software

Jan 2026 – June 2026

AI-powered JCL Modernization Platform

Hybrid • Pune, India

- Developed **15+** frontend and backend features using **Angular** and **FastAPI** for an AI-powered JCL modernization platform.
- Integrated an **Large Language Models (LLMs)** with manual user prompting, leveraging the compiler frontend to process **ASTs** for JCL explanation, syntax correction, and modernization.
- Designed and implemented a **multi-tab JCL editor** with persistent session storage, allowing users to seamlessly resume work across sessions.
- Integrated the **Topaz API** to securely communicate with IBM Mainframe systems for automated JCL retrieval and processing.

Tech Stack: *Angular, FastAPI, Python, JCL, AST, Compiler Frontend, LLM, Topaz API*

Projects

MiniLang IDE | [🌐 GitHub](#)

- Developed a full-stack IDE for a custom programming language using **React**, **Python**, and **ANTLR4**, enabling users to write, parse, visualize, and execute code.
- Designed a custom language grammar and implemented lexical analysis, parsing, AST generation, and an interpreter using the **Visitor Pattern**.
- Built interactive visualizations for tokens, Abstract Syntax Tree (AST), execution output, and memory state to simplify compiler concepts and program execution.
- Tech Stack:** React.js, Python, ANTLR4, AST, Visitor Pattern, Git

IPR Management System | [🌐 GitHub](#) | [🔗 Live Demo](#)

- Developed a full-stack web application to streamline the filing, tracking, and management of patents, trademarks, and copyrights.
- Implemented secure authentication, role-based access control, REST APIs, and centralized record management for Admin and Applicants.
- Deployed the **frontend on Vercel** and the **backend on Render**, using environment variables for secure configuration and seamless API communication.
- Tech Stack:** React.js, Vite, Node.js, Express.js, MongoDB, REST APIs, Vercel, Render, Git

Skills

Programming Languages: Java, Python, JavaScript, JCL, HTML/CSS, Bash

Frameworks & Libraries: Angular, React.js, FastAPI, Node.js, Express.js, Flask, Tailwind CSS

Databases: MongoDB, SQL, DB2

Tools & Platforms: Git, GitHub, REST APIs, Azure, AWS, Firebase, GitHub Actions, Topaz API, MCP (Model Context Protocol), ISPF

Certifications & Achievements

Postman API Fundamentals Student Expert – Postman

Mainframe Launchpad Certification – BMC Software

CCNA: Introduction to Networks – Cisco Networking Academy.

Received **Software Copyright Registration** from the Government of India for the **E-Voting System** project.